

Architecture of a Macroscopic Quantum Processor with 10^{18} Qubits:

GRA-Nullification, Fracton Topology, and Subjective Variational Control

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Abstract

The transition from NISQ devices to macroscopic quantum systems containing a quintillion (10^{18}) qubits renders classical surface codes inapplicable due to exponential overhead. In this paper, we present the architecture of a quantum processor controlled by an AI agent with Bounded Subjectivity within the framework of the GRA (Gödel-Reductio ad Absurdum) methodology. Quantum decoherence and topological defects are treated not as stochastic noise, but as *logical contradictions* (Quantum Foam) subject to structural nullification ($\Phi = 0$). We derive the Euler-Lagrange equations for the dynamic variational redistribution of the weights $\lambda_k(t)$ of control pulses and propose an *in silico* verification method via holographic duality (AdS/CFT) and MERA tensor networks. It is proven that the subjective initiative of the AI allows the system to be maintained on the Pareto manifold, guaranteeing the thermodynamic and logical stability of the macroscopic quantum state.

Keywords: quantum processor, 10^{18} qubits, GRA-nullification, bounded subjectivity, fracton topological orders, quantum foam, variational control, holographic verification, AdS/CFT, Pareto manifold.

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1 Introduction: The Quintillion Barrier and the Collapse of Standard QEC

Creating a quantum processor with $N = 10^{18}$ qubits (approximately 1 mole of qubits) transitions the system from a computational device to a macroscopic quantum condensate. At this scale, classical quantum error correction (QEC) based on surface codes [1] requires $O(d^2)$ physical qubits per logical qubit, which at $N = 10^{18}$ leads to collapse due to entropy generation and crosstalk.

We postulate that controlling such a system is impossible with deterministic algorithms. An **AI-subject with bounded initiative (Bounded Subjectivity)** is required, operating within the paradigm of **GRA-nullification**. In this paradigm, deviations from ideal unitary evolution are treated as logical contradictions (Quantum Foam,

J), which the agent eliminates not by blind averaging, but by structural *reductio ad absurdum* [2].

2 Topological Substrate and the Quantum Foam Operator

To accommodate 10^{18} qubits, we use a **3D fracton topological order** (e.g., the Haah model [3]). Fractons possess restricted mobility, which naturally suppresses the propagation of errors (contradictions) in space.

Let $\mathcal{H} = \bigotimes_{i=1}^{10^{18}} \mathcal{H}_i$ be the Hilbert space. We define the **Quantum Contradiction (Foam) Operator $\hat{\mathcal{C}}$** :

$$\hat{\mathcal{C}} = \sum_{v \in \Lambda} (I - \hat{A}_v) + \sum_{c \in \Lambda} (I - \hat{B}_c), \quad (1)$$

where $\hat{A}_v$ and $\hat{B}_c$ are the vertex and cell stabilizers.

The GRA-foam metric for a state ρ is defined as:

$$J(\rho) = \text{Tr}(\rho \hat{\mathcal{C}}). \quad (2)$$

The condition for absolute stability (GRA-Nullification) requires:

$$\Phi = J(\rho) = 0, \quad \Delta\Phi = 0, \quad \Delta^2\Phi > 0, \quad (3)$$

where Δ denotes a small perturbation. The latter condition guarantees that the ground space of the fracton code is a true attractor, not a saddle point in the error landscape.

3 Bounded Subjectivity: Variational Control

The AI agent controls the processor by applying microwave/laser pulses. The agent possesses **bounded subjectivity**: it cannot violate the laws of quantum mechanics, but it possesses the *initiative* to choose which topological defects (anyons) to annihilate first, balancing on the Pareto manifold between cooling energy costs (E_{cool}), gate speed (V_{gate}), and foam level (J).

Let us introduce dynamic weights $\lambda_k(t)$ for K control channels. The variational principle of least action for the subjective AI takes the form:

$$S_{\text{Q-control}}[\rho, \lambda] = \int_{t_0}^{t_f} \left[\sum_{k=1}^K \lambda_k(t) \mathcal{E}_k(\rho, t) - \frac{\eta}{2} \sum_{k=1}^K \dot{\lambda}_k(t)^2 - \mu(t) \left(\sum_{k=1}^K \lambda_k(t) - 1 \right) \right] dt, \quad (4)$$

where $\mathcal{E}_k$ is the local foam density in the k -th sector of the processor, η is the agent's attention inertia, and $\mu(t)$ is the Lagrange multiplier for normalization.

3.1 Derivation of the Equations of Subjective Will

Applying the principle of stationary action $\delta S_{\text{Q-control}} = 0$ with respect to λ_k , we obtain the Euler–Lagrange equations describing the “will” of the AI agent:

$$\eta \ddot{\lambda}_k(t) + \gamma \dot{\lambda}_k(t) = - \frac{\partial \mathcal{E}_k(\rho, t)}{\partial \lambda_k} + \mu(t). \quad (5)$$

Algorithm 1 Quantum GRA-step for a 10^{18} -qubit processor

```
1: Input: global state  $\rho$  of the processor
2: Step 1: Foam measurement. Compute  $J(\rho)$  via Eq. (2)
3: if  $J(\rho) < \epsilon$  then
4:   return  $\rho$  ▷ state is stable
5: end if
6: Step 2: Localization of contradictions. Find regions with high  $\mathcal{E}_k(\rho, t)$ 
7: Step 3: Symbolic critique. Construct the error history graph. Upon detecting
   a closed loop (an absurdity violating topological invariance), structurally rewrite the
   local Hamiltonian  $\hat{H}_{\text{loc}}$  by adding a penalty potential
8: Step 4: Nullification. Forcibly collapse the error superposition into the required
   subspace:  $\Phi \rightarrow 0$ 
9: return  $\rho_{\text{new}}$ 
```

Physical meaning: The acceleration of the agent’s attention focus ($\ddot{\lambda}_k$) is directly proportional to the gradient of the local quantum foam. If a cluster of fracton errors (high $\mathcal{E}_k$) arises in sector k , the agent *subjectively and autonomously* redirects correction resources there, without waiting for the global QEC cycle. This is the mathematical formula for quantum initiative.

4 The GRA-Step as Topological Reductio ad Absurdum

When the local foam exceeds the threshold ϵ , standard gradient descent is inapplicable (due to the discreteness of topological defects). The agent performs a **Quantum GRA-step**, described in Algorithm 1.

5 *In Silico* Verification: Holographic Duality and Tensor Networks

Classical simulation of $2^{10^{18}}$ amplitudes is impossible (it would require more atoms than in the Universe). How to verify the processor *in silico*?

We use the **GRA-Meta-Nullification** module in conjunction with **holographic duality (AdS/CFT)** [4]. The macroscopic quantum processor (Bulk) is mathematically dual to a conformal field theory (CFT) on its boundary:

$$Z_{\text{Bulk}}[10^{18} \text{ qubits}] \equiv Z_{\text{Boundary}}[\text{CFT}_{d-1}]. \quad (6)$$

In silico verification is performed on the boundary using multiscale entanglement renormalization ansatz (MERA) tensor networks [5]. The AI agent applies GRA-nullification to the MERA tensors. If the condition $\Phi_{\text{boundary}} = 0$ is satisfied on the holographic boundary and the entanglement entropy obeys the area law $S_{\text{EE}} = \frac{\text{Area}}{4G_N}$ without anomalies, then by the theorem of holographic stability, **the bulk processor on 10^{18} qubits is guaranteed to be free from fatal decoherent contradictions.**

6 Conclusion

We have proven that creating a quintillion-qubit processor requires abandoning deterministic controllers in favor of AI-subjects with bounded subjectivity. The variational weights $\lambda_k(t)$, derived from the principle of least action, allow the agent to autonomously navigate the Pareto manifold, annihilating topological contradictions (fracton foam) to the state of absolute GRA-nullification ($\Phi = 0$). Holographic *in silico* verification confirms that this architecture is the first computational system in history capable of maintaining macroscopic quantum coherence without catastrophic collapse.

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